# Supplemental Material for "Resolution refinement followed by the rodeo algorithm: a gate-level benchmark of eigenstate-preparation routes on the pairing Hamiltonian"

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This Supplemental Material collects the verification material behind the main text: the comparison of coarse solvers at larger basis size (Sec. S1), the refinement energies and the $k = 3 \to 4$ chain (Sec. S2), the shift-scan fits and the response norm along the path (Sec. S3), the evaluation of the sufficient adiabatic-time and Trotter-step conditions (Sec. S4), the rodeo energy scan and its statistics (Sec. S5), the operating points behind the fidelities quoted in the main text (Sec. S6), the other model-space sizes, particle numbers and observables (Sec. S7), the validation of the pair-basis implementation with the ingredients of the sixteen-qubit study (Sec. S8), and the convergence of the operating-point search behind Table V of the main text (Sec. S9). Equation, section, figure and table numbers without the prefix S refer to the main text. All numbers are produced by the notebook of the code repository.

## S1. COARSE SOLVERS AT LARGER BASIS SIZE

Figure S1 compares all coarse solvers of Sec. IV A of the main text — exact diagonalization, coupled-cluster doubles, statevector and gate-level UCCSD, and a hardware-efficient ansatz — as a function of the number of levels $k$ at $N = 4$, $g = 1$, and Fig. S2 follows the gate-level UCCSD solver to $k = 8$ (sixteen qubits). The hardware-efficient ansatz is exact at $k = 2$, reaches FCI at $k = 3$ only after a multi-start optimization with a particle-number penalty, and fails at $k = 4$; the UCCSD solver stays variational and tight throughout, at a CNOT count that grows steeply (1 312 at $k = 4$).

*a. Hardware-efficient ansatz.* The HEA [1] used for contrast consists of $L$ layers of $R_y$ rotations on all $2k$ qubits interleaved with a CNOT entangler (nearest-neighbor pairs plus a wrap-around bond), i.e. $2k(L+1)$ variational angles and only $O(2kL)$ CNOTs: 12 angles at $k = 2$ ($L = 2$) and 30 at $k = 3$ ($L = 4$). Its depth is far smaller than UCCSD, but the circuit carries no particle-number or seniority structure; a number-penalty term and multi-start optimization are needed at $k = 3$, and at $k = 4$ the optimizer fails to reach the ground state (Fig. S1). Depth is not the only currency: trainability of the symmetry-blind landscape becomes the bottleneck first.

## S2. REFINEMENT ENERGIES AND THE $k = 3 \to 4$ CHAIN

Figure S3 shows the overlap and the physical energy $\langle\Phi(T)|H_{\rm high}|\Phi(T)\rangle$ of the refined state versus the adiabatic time $T$ for the $k = 2 \to 3$ and $2 \to 4$ chains at $N = 4$, $g = 1$ (Sec. IV B of the main text), and Fig. S4 compares the $2 \to 4$ chain with the $3 \to 4$ chain that starts from the gate-level UCCSD state at $k = 3$ (Sec. IV C).

## S3. SHIFT SCAN: FITS AND RESPONSE NORM

Figure S5 shows the two affine fits to the four shift-scan points of Fig. 4 of the main text, in $1/\Delta E_{\rm min}$ and in $1/\Delta E^2_{\rm min}$, with their sums of squared residuals (28 and 0.2), and the first-order response norm $\chi_u(s)$ along the $k = 2 \to 4$ path for the four shifts; at $s = 0$ it equals $g^2/b^2$ with $b = \mu - E_{\rm low}$, the $k = 4$, $N = 4$ case of the identity $\chi_u(0) = \|QH_{\rm high}P\psi_{\rm low}\|^2/b^2$, Eq. (13) of the main text, which gives $P(k-P)g^2/(4b^2)$ for the filled pairing seed.

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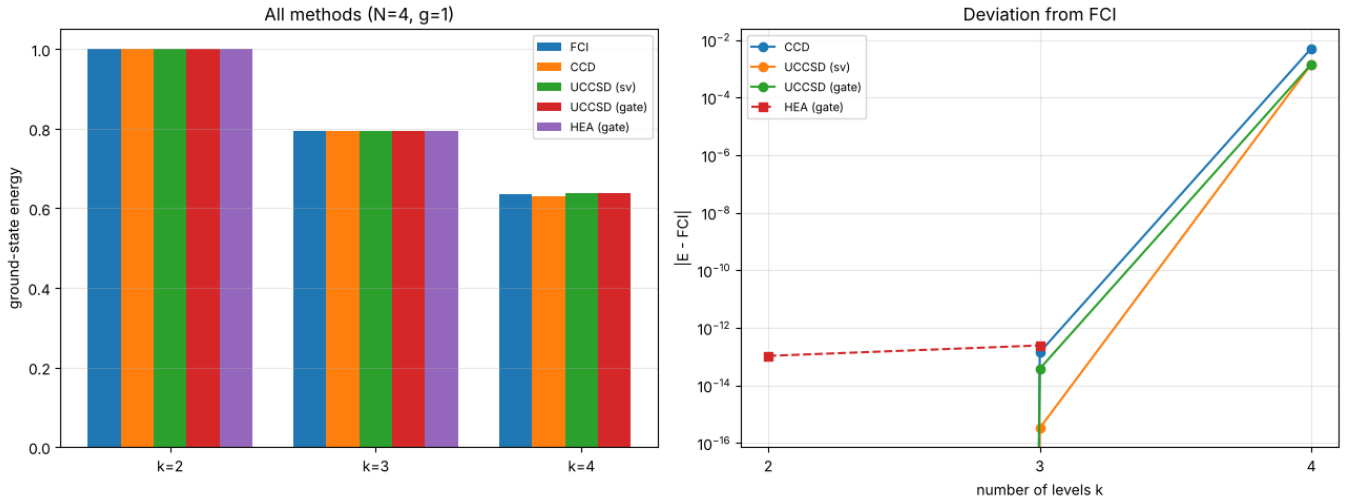



FIG. S1. Ground-state energy at $N = 4$, $g = 1$ obtained with all solvers (left) and their deviation from FCI (right), versus the number of levels $k$. The structured UCCSD circuits track FCI from above; CCD over-binds; the hardware-efficient ansatz matches FCI for $k \leq 3$ but loses accuracy at $k = 4$.

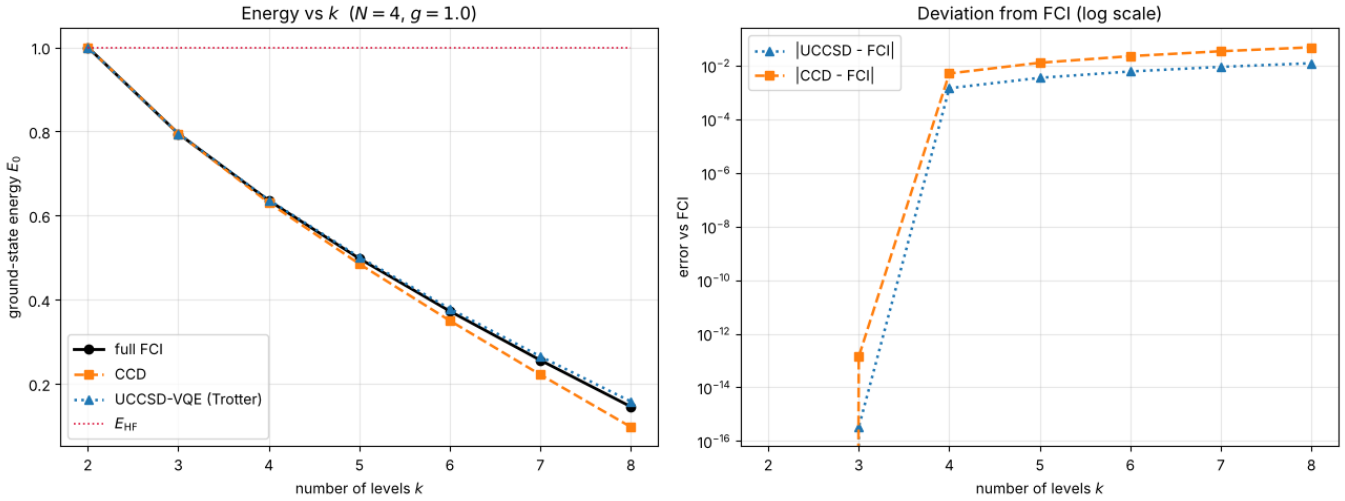



FIG. S2. Scalability of the Trotterized UCCSD–VQE: ground-state energy (left) and deviation from FCI (right) from $k = 2$ to $k = 8$ ($N = 4$, $g = 1$). The same solver remains variational and tight as the basis — and hence the qubit count — grows.

## S4. SUFFICIENT CONDITIONS VERSUS PRACTICE

The conditions quoted in Sec. III C of the main text can be evaluated exactly for the path of Eq. (3). Writing $A = P(H_{\text{low}} - \mu)P^\dagger$ and $B = H_{\text{high}} - \mu$, the schedule gives $\dot{H} = \frac{\pi}{2}\sin(\pi s)\,(B-A)$ and $\ddot{H} = \frac{\pi^2}{2}\cos(\pi s)\,(B-A)$, so the integrand of Eq. (6) is fixed by the single number $\|B - A\|$ and by the instantaneous gap $\Delta(s)$, both taken in the $N = 4$ sector in which the evolution takes place. Figure S6 shows the two along the $k = 2 \to 4$ path: the gap rises from the shift-controlled 0.600 at $s = 0$ to the physical gap of $H_{\text{high}}$ at $s = 1$, and the integrand peaks near $s \simeq 0.25$, where $\sin(\pi s)$ is already sizable while the gap is still close to its minimum; the term $7\left\|\dot{H}\right\|^2/\Delta^3$ supplies 94% of the integral. Table S1 collects the numbers. Here $\|B - A\| = 7.62$ and $\int_0^1 [\cdots]\, ds = 511$, so Eq. (6) demands $T \geq 5.1 \times 10^3$ for an infidelity of $10^{-2}$ (and $1.6 \times 10^4$ for $10^{-3}$), whereas the computed overlap reaches 0.99 at $T^* = 16$: the sufficient condition overshoots the practical time by a factor $3 \times 10^2$. The step-count condition is looser still. With $\Lambda_1 = \max_s \left\|\dot{H}\right\| = 12.0$ and, for the term partition of the gate compilation of Sec. V B of the main text (the conditioned coarse term, the diagonal term and the six pair-hopping terms), $\Lambda_2 = \max_s \sum_{i<j} \|[H_i(s), H_j(s)]\| = 13.0$, the first-order bound at $\eta = \sqrt{\epsilon}/2$ requires $3.4 \times 10^4$ steps along the

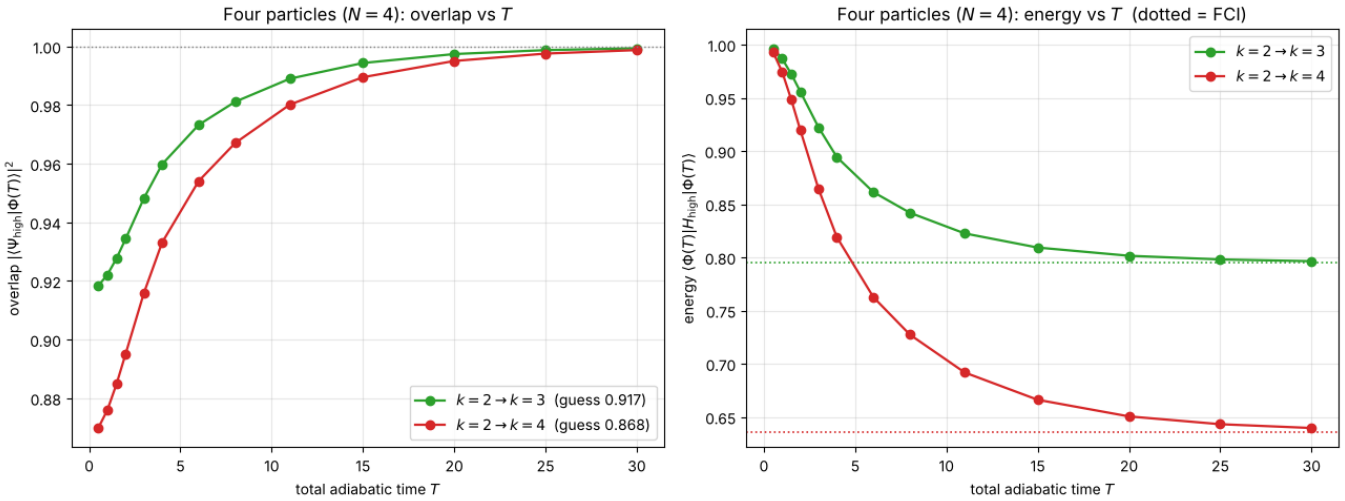



FIG. S3. Four-particle refinement up to $k = 4$: overlap (left) and physical energy $\langle\Phi(T)|H_{\rm high}|\Phi(T)\rangle$ (right) versus $T$ for $k_{\rm high} = 3, 4$. The energy approaches the FCI value (dotted) from above while the overlap approaches unity (the time scale is empirical; the scan of Sec. IV D of the main text identifies no exponent).

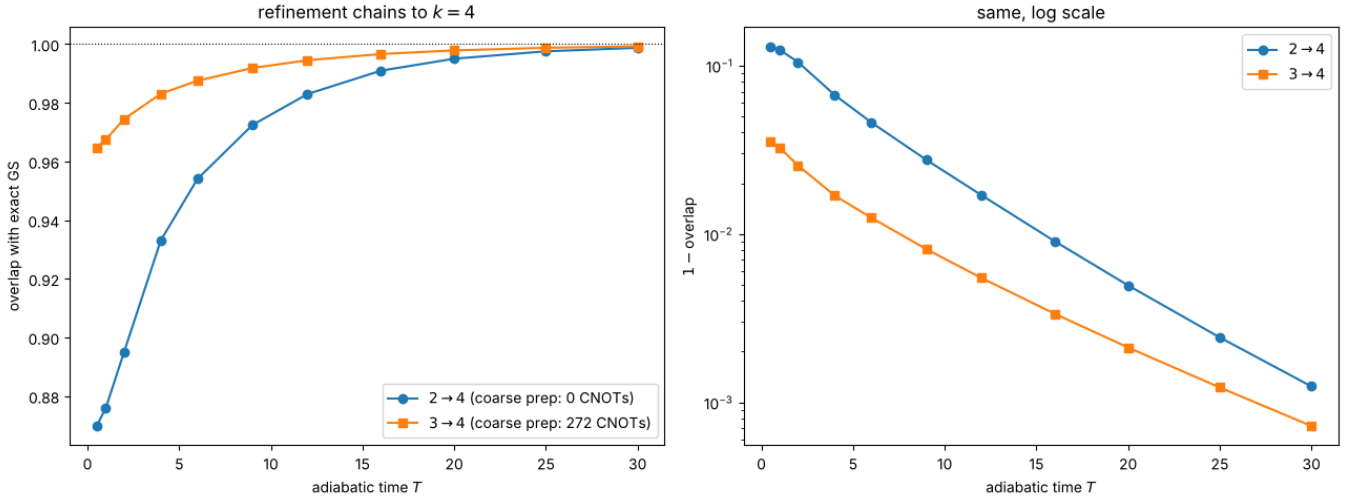



FIG. S4. Refinement chains to $k = 4$ at $N = 4$, $g = 1$, starting from the gate-free $k = 2$ determinant (circles) and from the 272-CNOT gate-level UCCSD state at $k = 3$ (squares). Left: overlap with the exact ground state versus adiabatic time $T$; right: the same on a logarithmic scale. The $3 \to 4$ chain starts at overlap 0.964 (versus 0.868) and reaches 0.99 in half the adiabatic time ($T^* = 8$ on a finer grid than the one shown).

$T^* = 16$ path — and $3.4 \times 10^9$ along the path of length $T_{\rm b}$ — while 160 first-order or 40 second-order steps already reproduce the exact-slice overlap at $T^*$ to within the adiabatic error of the path itself. The two step counts answer different questions and the ratio between them must be read with that in mind: the bound certifies an operator-norm error of the digitized unitary, whereas reproducing one scalar overlap does not certify that the two output vectors, let alone the two unitaries, are close.

These ratios — $3 \times 10^2$ for the time and $2 \times 10^2$ for the step count — are the fermionic counterpart of what Ref. [3] reports for nucleation of a twelve-site Ising chain ($6 \times 10^1$ and $10^7$, respectively), and they carry the same message: the conditions certify convergence but do not predict cost. Part of the reason is visible in the integrand. The bound pays for the full operator norm $\|B - A\| = 7.62$ — the same value in the full $N = 4$ sector and in the six-dimensional seniority-zero block that the dynamics actually explores, so it is not an artifact of inaccessible sectors — and for the global gap, whereas the adiabatic error is governed by the individual matrix elements of $dH/ds$ between the tracked state and the states it couples to, relative to their individual gaps. Those elements are small but not zero for the highest accessible states: at $s = 0.25$ the derivative couplings from the ground state to the five seniority-zero excited states are 0.60, 0.37, 0.00,

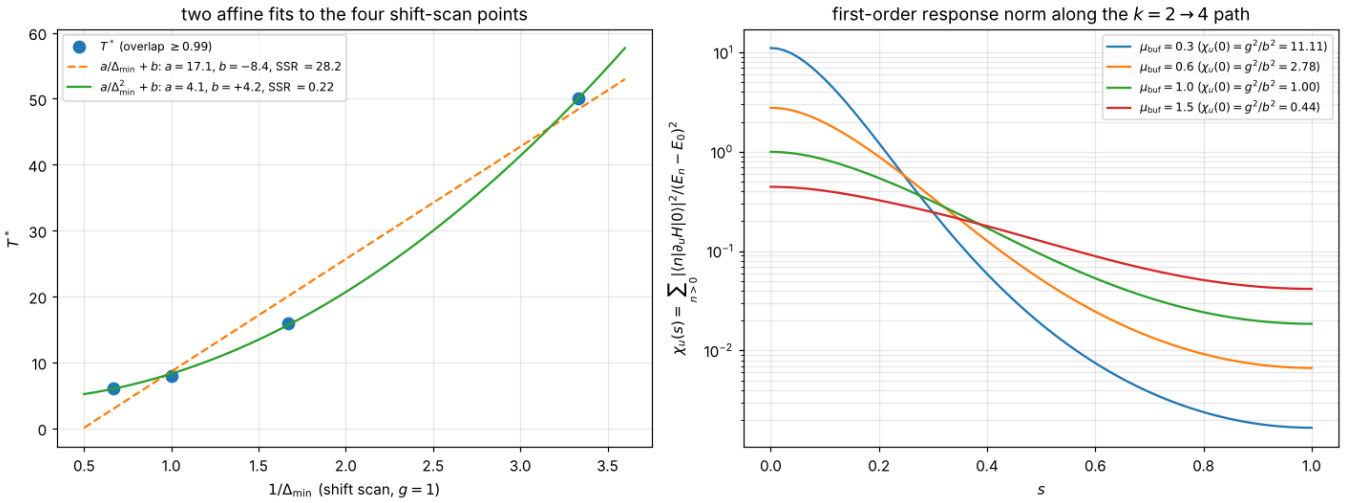



FIG. S5. Left: the four shift-scan points of Fig. 4 with the two affine fits, in $1/\Delta E_{\min}$ (dashed) and in $1/\Delta E^2_{\min}$ (solid), and their sums of squared residuals. Right: the first-order response norm $\chi_u(s) = \sum_{n>0} |\langle n|\partial_u H|0\rangle|^2/(E_n - E_0)^2$ of the $k = 2 \to 4$ path, $u = \sin^2(\pi s/2)$, for the four shifts; at $s = 0$ it equals $g^2/b^2$ with $b = \mu - E_{\text{low}}$ and it is largest there, so the quantity that the inverse-gap argument assumes bounded changes by a factor 25 across the scan ($\chi_u$ is the response along the affine path of Ref. [2]; the response to the scheduled parameter, $\chi_s = [\frac{\pi}{2} \sin \pi s]^2 \chi_u$, vanishes at both ends and its maximum varies by a factor 3.4).

0.11 and 0.08. A global norm therefore overestimates the cost by ignoring the state-specific structure of the couplings; the magnitude of the overshoot alone does not single out one cause, nor does it verify the inverse-gap mechanism of Ref. [2].

## S5. THE RODEO ENERGY SCAN

Figure S7 shows the energy scans of Sec. IV E of the main text ($M = 4$ cycles, 12 time sets per energy). The coarse scan ($\sigma = 2$, 60 energies on $[-0.5, 4]$) resolves the ground-state peak from the excited band; the fine scan ($\sigma = 8$, 161 energies with spacing 0.0075 on a window of width 1.2 centered on the highest point of the coarse scan, $E = 0.644$, i.e. chosen without knowledge of $E_0$) locates the peak at $E = 0.6362$ against the exact $E_0 = 0.635548$ by three-point parabolic interpolation at the grid maximum. On a device every acceptance is a shot estimate: adding binomial noise for $10^3$ and $10^4$ shots per energy and locating the peak by a quadratic fit through the points above 80% of the maximum gives $0.637 \pm 0.001$ and $0.6373 \pm 0.0004$, so with this scan the systematic error of the peak estimator ($\sim 2 \times 10^{-3}$, from the non-parabolic shape of the peak over the fitted window) exceeds the shot noise, and the agreement to $10^{-3}$ is a property of the scan design rather than a demonstrated four-decimal precision. Starting from the refined state, essentially all spectral weight sits in the single ground-state peak; starting from the prolonged determinant, the small residual weight on excited states is visible as secondary structure above the $2^{-M}$ background. The fine scan alone evaluates 161 energies with 12 time sets each, about $2 \times 10^3$ four-cycle runs, each repeated over enough shots to estimate an acceptance probability.

## S6. OPERATING POINTS

Table S2 records the operating points behind the different fidelities quoted in the main text, which belong to different stages of idealization: the ideal filter of Sec. IV E and the digital pipeline of Secs. V B–V C differ in refinement time, slice treatment, filter width and evolution, and the residual errors of the former are not achieved by the counted circuits of the latter.

## S7. OTHER MODEL-SPACE SIZES, PARTICLE NUMBERS, AND OBSERVABLES

To see whether the ranking of Table II of the main text is an accident of the smallest nontrivial case, Table S3 repeats the comparison, with exact-slice refinement and exact controlled evolution ($\sigma = 3$, 40 time sets), for $N = 4$ and $N = 6$ particles on $k = 4$ and $k = 6$ levels (eight and twelve qubits), always starting from the filled determinant of the smallest coarse space ($k_{\text{low}} = N/2$). The pattern is the same throughout: the prolonged overlap falls to 0.78–0.80 at $k = 6$, refinement for $T = 30$ restores it to 0.997–0.999, and the UCCSD input, whose infidelity grows from $1.8 \times 10^{-4}$ to $1.0 \times 10^{-3}$ over the table, still enters the filter with a higher overlap than the refined state and leaves it with a lower infidelity at every cycle number. (At $N = 6$, $k = 4$ the ansatz is exact: three pairs on four levels are the particle–hole conjugate

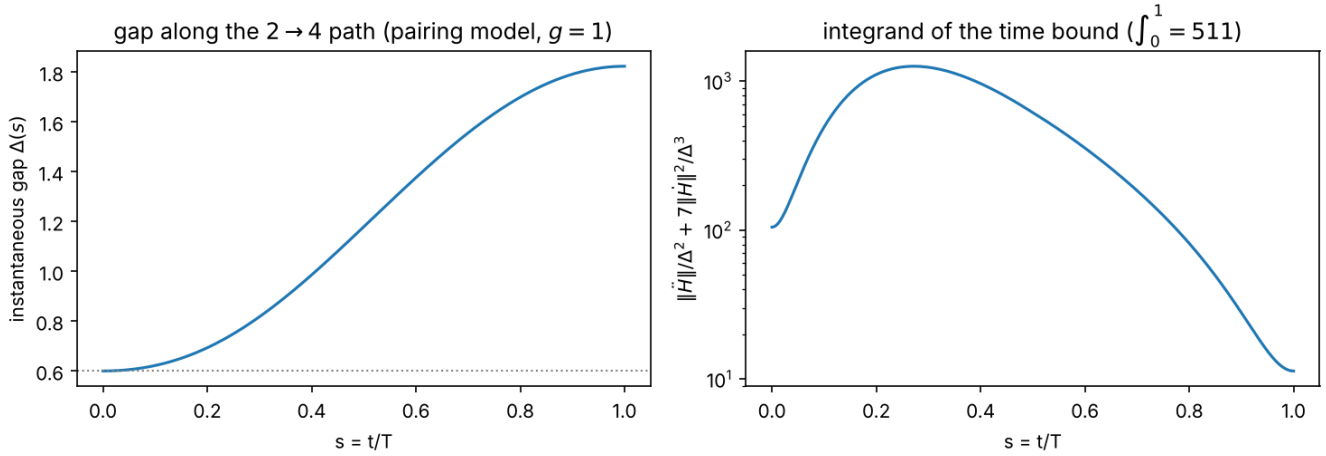



FIG. S6. The ingredients of the adiabatic-time bound, Eq. (6), along the $k = 2 \to 4$ refinement path at $N = 4$, $g = 1$, $\mu = E_{\rm low} + 0.6$. Left: instantaneous gap $\Delta(s)$ in the $N = 4$ sector, from the shift-controlled 0.600 at $s = 0$ (dotted line) to the physical gap 1.823 of $H_{\rm high}$ at $s = 1$. Right: the integrand $\left\|\ddot{H}\right\|/\Delta^2 + 7\left\|\dot{H}\right\|^2/\Delta^3$ (logarithmic scale), which peaks near $s \simeq 0.25$ and integrates to $I = 511$ (Table S1).

TABLE S1. Sufficient conditions of Ref. [3] evaluated for the $k = 2 \to 4$ refinement path ($N = 4$, $g = 1$, $\mu = E_{\rm low} + 0.6$; spectral norms in the $N = 4$ sector) against the empirical cost. $\Delta_{\rm min}$: minimum instantaneous gap; $I$: the integral in Eq. (6); $T_{\rm b}$: the time bound for infidelity $\epsilon = 10^{-2}$, $T_{\rm b} = I/\sqrt{\epsilon}$; $T^*$: time for overlap 0.99 (grid of Fig. 4); $\Lambda_1 = \max_s \left\|\dot{H}\right\|$ and $\Lambda_2 = \max_s \sum_{i<j} \|[H_i, H_j]\|$ over the Trotter groups of the gate compilation; $n_{\rm b}(T^*)$: first-order Trotter steps required by the bound along the $T^*$ path at $\eta = \sqrt{\epsilon}/2$; $n_1$, $n_2$: first- and second-order steps (from the sequence $5, 10, 20, \ldots$) that reproduce the exact-slice overlap at $T^*$ to within the adiabatic error $1 - p(T^*)$ of the path.

| $\Delta_{\rm min}$ | $\|B - A\|$ | $I$ | $T_{\rm b}$ | $T^*$ | $T_{\rm b}/T^*$ | $\Lambda_1$ | $\Lambda_2$ | $n_{\rm b}(T^*)$ | $n_1$ | $n_2$ |
|---|---|---|---|---|---|---|---|---|---|---|
| 0.600 | 7.62 | 511 | $5.1 \times 10^3$ | 16 | $3.2 \times 10^2$ | 12.0 | 13.0 | $3.4 \times 10^4$ | 160 | 40 |

of one pair, for which UCCSD is complete.) The price of the UCCSD input is its circuit — 6 976 CNOTs and 92 parameters at $k = 6$, $N = 4$; 9 024 CNOTs and 117 parameters at $k = 6$, $N = 6$ — and the training of those parameters, which the refinement avoids at the cost of $2n_s(N^{\rm step}_{\rm CNOT} + N^A_{\rm CNOT})$ CNOTs with $N^{\rm step}_{\rm CNOT} = 732$ at $k = 6$. This is a controlled small-system demonstration, not a scaling study; a regime in which refinement wins over the single-layer variational input requires a space for which that input is poor, and Sec. V E of the main text shows that four pairs on eight levels is already such a space (the adaptively grown ansatz remains far cheaper there).

*a. Observables.* A state-preparation method for nuclear structure is judged by the observables of the state it delivers, not only by its overlap. Table S4 lists the level occupation numbers $n_p = \sum_\sigma \langle n_{p\sigma} \rangle$ and the correlation energy $E - E_{\rm ref}$ of the states of Table II of the main text at $k = 4$, $N = 4$, $g = 1$ (for the filtered states, the acceptance-weighted average over the 40 time sets). The exact ground state depletes the two occupied levels by 0.07 and 0.20 particles. The UCCSD state reproduces the occupations to $3 \times 10^{-3}$ and the correlation energy to $1.4 \times 10^{-3}$; the refined state at $T = 30$, whose infidelity is $2.3 \times 10^{-3}$, deviates by up to 0.02 in the occupations and by $10^{-2}$ in the correlation energy, and two rodeo cycles reduce both deviations by a factor 2–3, six cycles by a factor 4–5. Occupation numbers converge more slowly than the energy, as expected for a quantity that is linear rather than quadratic in the error of the state.

## S8. THE PAIR BASIS AND THE SIXTEEN-QUBIT INGREDIENTS

Every stage of the pipeline conserves seniority — the Hamiltonian, the embedding, the path, the exactly exponentiated pair-hopping blocks and the filter — so for an $N$-paired target the whole computation can be carried out in the pair basis (one bit per level, $\binom{k}{N/2}$ states). The pair-basis implementation used for Sec. V E of the main text was validated against the full-sector code at $k = 4$, $N = 4$: the Hamiltonian and every Trotter term coincide with the seniority-zero blocks of their full-sector counterparts to machine precision, and every anchor of the main text is reproduced — the ground-

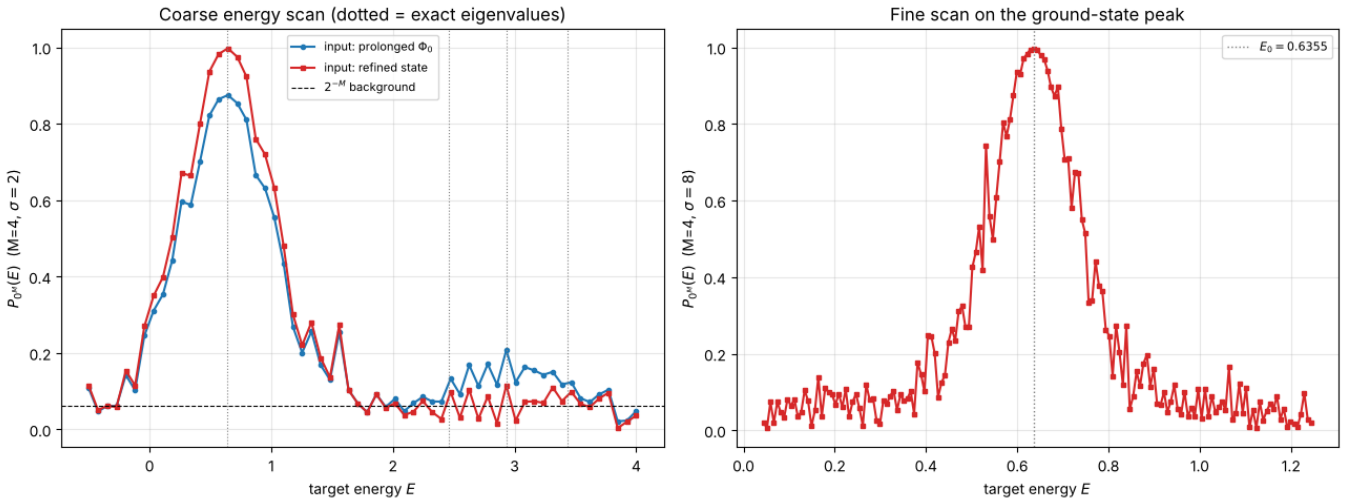



FIG. S7. Rodeo energy scan ($M = 4$). Left: coarse scan ($\sigma = 2$); the dotted vertical lines mark exact eigenvalues and the dashed line the $2^{-M}$ background. Right: fine scan ($\sigma = 8$) on a window centered on the coarse-scan maximum; the peak is located at $E = 0.6362$ (exact $E_0 = 0.6355$).

TABLE S2. Operating points behind the fidelities quoted in this paper ($k = 4$, $N = 4$, $g = 1$).

| result | assumptions and operating point | status |
|---|---|---|
| coarse state (Sec. III A) | factorized UCCSD, numerically optimized | valid circuit family; $\epsilon_{\text{low}}$ to be supplied or bounded |
| prolongation (Sec. IV B) | nested bare Hamiltonians, empty appended levels | gate-free embedding; expectation-value identity exact |
| adiabatic guarantee, Eq. (6) | exact coarse eigenstate, isolated branch, ideal continuous evolution | conditional bound, not the complete digital budget |
| refinement (Sec. IV B) | $T = 25$ or 30, exact midpoint slices (160) | ideal; slice error $\sim 10^{-6}$ |
| filter (Sec. IV E) | $T = 25$ refined input, $\sigma = 3$, exact controlled evolution, 40 schedules | ideal; sampling error up to 80% at $M = 10$ |
| circuit comparison (Secs. V B–V C) | product formulas, $T = 16$ or 30, $\sigma = 4$, step ceiling $\delta t = 0.25$ | digital; floor $1.1 \times 10^{-4}$ (converged averaged map), acceptance decaying 0.2% per cycle; sampled 40-schedule values in Table II, exact-channel minima in Table V |

state energy 0.6355484736, the prolonged overlap $p_0 = 0.8676375677$, the refined overlap $p(25) = 0.9975712826$, the Trotterized $p = 0.9922$ at $T^* = 16$ with 43 second-order steps, the four-pair ansatz $E = 0.6369869810$ with $1 - F = 1.776 \times 10^{-4}$, and the exact-ensemble infidelities $1.216 \times 10^{-3}$, $6.08 \times 10^{-4}$, $3.80 \times 10^{-5}$ and $2.38 \times 10^{-6}$ after 1, 2, 6 and 10 cycles. The gate counts of the sixteen-qubit study are those of the $2k$-qubit compilation of Sec. V of the main text.

Table S5 lists the ingredients of the sixteen-qubit study for $g = 0.5$ to 2, and Fig. S8 the exact-slice transport along the refinement chains $k_{\text{low}} \to 8$ and along the HF ramp, with its coupling dependence.

## S9. CONVERGENCE OF THE OPERATING-POINT SEARCH

The minimum costs of Table V of the main text are the best costs found under a stated finite search, not certified optima, and this section records how they move as the search is refined. Three levels are compared for the two adiabatic routes. L0 is the coarse grid of the first joint search (notebook, Sec. 22): $T$ in steps of 2, three shifts, eight step counts. L1 is a fine grid of the evolution time, $T = 3$ to 12 in steps of 0.25 for the ramp and 3 to 16 in steps of 0.5 for the refinement at $k = 4$, 3.5 to 10 in steps of 0.25 and 3.5 to 12 in steps of 0.5 at $k = 8$ and $g = 1$, 5 to 16 in steps of 0.5 for both at $g = 2$, with the shift $\mu_{\text{buf}}$ on ten values from 0.6 to 8 at $k = 4$, nine from 1 to 16 at $g = 1$ and nine from 4 to 32 at $g = 2$, and the second-order step count on a geometric list of ratio 1.15 from 6 to 160 ($k = 4$) or 16 to 400 ($k = 8$);

TABLE S3. The pipeline and the UCCSD comparator at other sizes ($g = 1$): coarse space the filled determinant at $k_{\rm low} = N/2$, exact-slice refinement for $T = 30$, exact controlled evolution with $\sigma = 3$, 40 time sets. $\mathrm{ov}_0$ and $p$ are the overlaps of the prolonged and refined states with the exact ground state, the UCCSD columns give the infidelity and energy error of the sector-level UCCSD state together with its parameter and CNOT counts, and the last four columns the acceptance-weighted infidelity after $M$ rodeo cycles from the refined and from the UCCSD input. The register has $2k$ qubits (8 and 12) and the uncontrolled first-order Trotter step of Eq. (11) costs 296 CNOTs at $k = 4$ and 732 at $k = 6$.

| $(k, N)$ | dim | gap | $\mathrm{ov}_0$ | $p\,(T{=}30)$ | $1 - \mathcal{F}_{\rm UCCSD}$ | $\Delta E_{\rm UCCSD}$ | params | CNOTs | refined+rodeo | | UCCSD+rodeo | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | $M{=}2$ | $M{=}6$ | $M{=}2$ | $M{=}6$ |
| $(4, 4)$ | 70 | 1.823 | 0.868 | 0.9988 | $1.8 \times 10^{-4}$ | $1.4 \times 10^{-3}$ | 26 | 1 312 | $3.4 \times 10^{-4}$ | $2.3 \times 10^{-5}$ | $4.1 \times 10^{-5}$ | $1.4 \times 10^{-6}$ |
| $(6, 4)$ | 495 | 2.017 | 0.798 | 0.9976 | $7.6 \times 10^{-4}$ | $6.1 \times 10^{-3}$ | 92 | 6 976 | $6.3 \times 10^{-4}$ | $4.1 \times 10^{-5}$ | $1.7 \times 10^{-4}$ | $7.6 \times 10^{-6}$ |
| $(4, 6)$ | 28 | 1.779 | 0.894 | 0.9991 | $< 10^{-15}$ | $< 10^{-15}$ | 15 | 688 | $2.5 \times 10^{-4}$ | $1.7 \times 10^{-5}$ | 0 | 0 |
| $(6, 6)$ | 924 | 2.052 | 0.780 | 0.9972 | $1.0 \times 10^{-3}$ | $8.3 \times 10^{-3}$ | 117 | 9 024 | $7.2 \times 10^{-4}$ | $4.8 \times 10^{-5}$ | $2.4 \times 10^{-4}$ | $1.0 \times 10^{-5}$ |

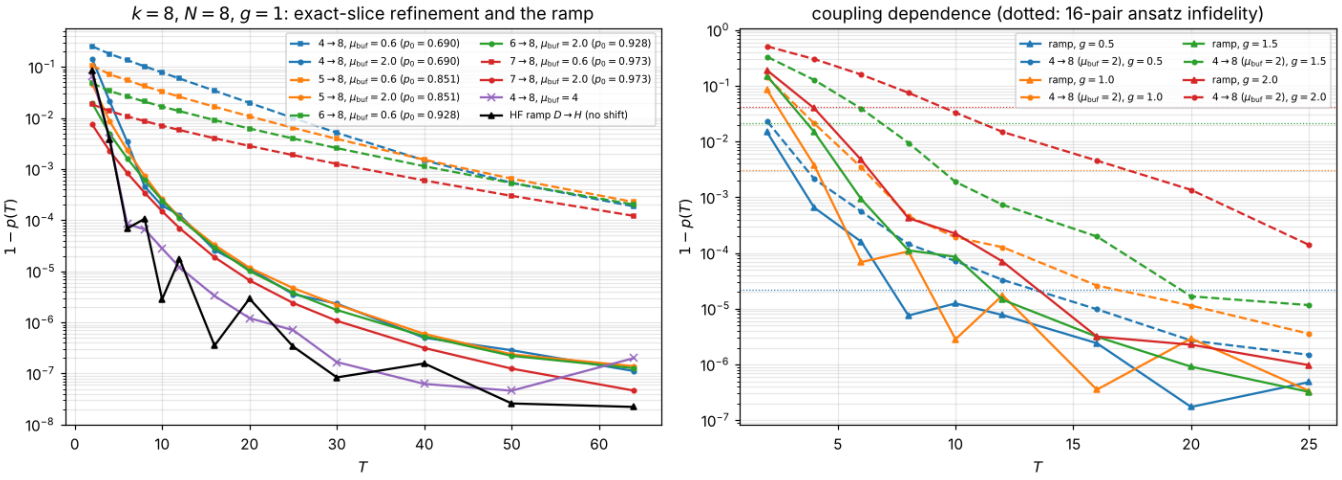



FIG. S8. Four pairs on eight levels ($N = 8$, $k = 8$, sixteen qubits), exact slices. Left, $g = 1$: $1 - p(T)$ for the refinement chains $k_{\rm low} \to 8$ with $k_{\rm low} = 4$–7 at $\mu_{\rm buf} = 0.6$ (dashed) and 2 (solid), the $4 \to 8$ chain at $\mu_{\rm buf} = 4$, and the HF adiabatic ramp from the same determinant. Right: coupling dependence of the ramp and of the $4 \to 8$ chain at $\mu_{\rm buf} = 2$ for $g = 0.5$, 1, 1.5 and 2; the dotted lines are the infidelities of the 16-parameter pair-double ansatz.

TABLE S4. Level occupation numbers $n_p$ and energy of the prepared states at $k = 4$, $N = 4$, $g = 1$ ($E_{\rm ref} = 1$); the filtered states use the Trotterized filter of Table II ($\delta t = 0.25$) and are acceptance-weighted averages over 40 time sets; $a\,10^{-b}$ stands for $a \times 10^{-b}$.

| state | $n_0$ | $n_1$ | $n_2$ | $n_3$ | $E - E_{\rm ref}$ | $1 - \mathcal{F}$ |
|---|---|---|---|---|---|---|
| exact | 1.929 | 1.803 | 0.197 | 0.071 | −0.3645 | 0 |
| reference det. | 2 | 2 | 0 | 0 | 0 | $1.3\,10^{-1}$ |
| UCCSD | 1.930 | 1.806 | 0.194 | 0.070 | −0.3630 | $1.8\,10^{-4}$ |
| refined ($T{=}30$) | 1.908 | 1.795 | 0.205 | 0.092 | −0.3549 | $2.3\,10^{-3}$ |
| refined+rodeo $M{=}2$ | 1.920 | 1.798 | 0.202 | 0.080 | −0.3621 | $5.3\,10^{-4}$ |
| refined+rodeo $M{=}6$ | 1.923 | 1.801 | 0.199 | 0.077 | −0.3637 | $1.5\,10^{-4}$ |
| UCCSD+rodeo $M{=}2$ | 1.924 | 1.801 | 0.199 | 0.076 | −0.3637 | $1.3\,10^{-4}$ |

every point is evaluated with the exact averaged channel for $M = 0, \ldots, 6$. L2 refines, for every control whose L1 cost lies within 40% of the L1 minimum, the integer step count by bracketing and bisection at fixed $(T, \mu_{\rm buf}, M)$. Table S6 lists the three levels, the operating point of L2, the number of distinct controls ($T$) or $(\mu_{\rm buf}, T)$ whose refined cost lies within 10% of the L2 minimum, and the ratio of the two routes. Figure S9 shows the L0 grid itself, with the exact-channel tracks of the fixed inputs of Table VI; the frontier curves of Fig. 9 of the main text are read from the L1 cloud.

Three features of Table S6 carry the conclusions drawn in the main text. First, the coarse grid was not converged: every adiabatic minimum falls on refinement, by 31–51% at $k = 4$, 25–40% at $k = 8$, $g = 1$, and 8–21% at $g = 2$, and the operating points move to shorter times ($T = 4.25$–$5.5$ instead of 4–8 at $g = 1$) and, for the refinement, to different shifts. Spot checks at intermediate times that were not on the coarse grid — the eight-qubit ramp at $T = 4.5$, $n_s = 11$ ($1.63 \times 10^{-3}$ for 6 512 CNOTs) and at $T = 4.75$, $n_s = 16$ and 22; the sixteen-qubit ramp at $T = 5$, $n_s = 40$ and $T = 5.2$, $n_s = 60$; the $g = 2$ refinement at $\mu_{\rm buf} = 16$, $T = 10$, $n_s = 160$ — all lie on or near the fine-grid frontier. Second, the fine grid is close to converged in the sense available to a finite search: the integer refinement L1→L2 moves the min-

TABLE S5. Four pairs on eight levels ($N = 8$, $k = 8$): HF and exact energies, seniority-zero gap, HF overlap $p_{\mathrm{HF}}$, the 16-parameter pair-double ansatz (768 CNOTs) and the 8-parameter coarse ansatz at $k = 6$ (384 CNOTs) with the overlap of its prolongation, the response norm $\chi_u(0) = P(k-P)g^2/(4b^2)$ at $b = 0.6$, and the exact-slice $1-p$ at $T = 8$ of the HF ramp, of the $4 \to 8$ chain ($\mu_{\mathrm{buf}} = 2$) and of the $6 \to 8$ chain from the exact coarse state ($\mu_{\mathrm{buf}} = 2$). The shift scan of the $4 \to 8$ chain is Table IV of the main text.

| $g$ | $E_{\mathrm{HF}}$ | $E_0$ | gap | $p_{\mathrm{HF}}$ | $1-F_{16}$ | $1-F_8^{k=6}$ | $p_0(6\to8)$ | $\chi_u(0)$ | $1-p(T{=}8)$: ramp / 4→8 / 6→8 |
|---|---|---|---|---|---|---|---|---|---|
| 0.5 | 11 | 10.790 | 2.06 | 0.940 | $2.2\times10^{-5}$ | $7.3\times10^{-6}$ | 0.989 | 2.8 | $7.6\times10^{-6}$ / $1.4\times10^{-4}$ / $1.1\times10^{-4}$ |
| 1 | 10 | 8.889 | 2.59 | 0.690 | $3.1\times10^{-3}$ | $6.9\times10^{-4}$ | 0.928 | 11.1 | $1.1\times10^{-4}$ / $4.6\times10^{-4}$ / $6.1\times10^{-4}$ |
| 1.5 | 9 | 6.055 | 3.97 | 0.413 | $2.1\times10^{-2}$ | $5.3\times10^{-3}$ | 0.817 | 25.0 | $1.1\times10^{-4}$ / $9.5\times10^{-3}$ / $1.9\times10^{-3}$ |
| 2 | 8 | 2.487 | 5.95 | 0.253 | $4.2\times10^{-2}$ | $1.3\times10^{-2}$ | 0.709 | 44.4 | $4.2\times10^{-4}$ / $7.5\times10^{-2}$ / $4.7\times10^{-3}$ |

ima by 0–12%, and the frontier is sharp in $T$ at $g = 1$ (one to six controls within 10% of the minimum) because the cheapest operating point sits at the shortest time at which the adiabatic error alone meets the target, after which the step count is fixed by the digitization error; at $g = 2$ the frontier is flatter (two to thirteen controls within 10%), the ramp needs one and three filter cycles at the two tighter targets, and the refinement reaches all three unfiltered. A finer grid in $T$ (steps of 0.05 around the $g = 1$ ramp minima at $10^{-3}$ and $3\times10^{-4}$) returned the same two minima, $7.07\times10^4$ and $9.79\times10^4$ CNOTs; we nevertheless do not interpret differences between the two routes below the 10% level, which is the size of the L1→L2 step. Third, the ratio of the two routes changes sign with the coupling: 0.85–0.95 at $k = 4$, 1.22–1.42 at $k = 8$, $g = 1$, and 0.86–1.01 at $g = 2$. The 19% overhead of the conditioned coarse term (254 of 1 614 CNOTs per step) is therefore not the whole separation at $g = 1$ — the shifted path also needs a somewhat longer time at the tighter targets — and at $g = 2$ it is more than compensated by the cycles the ramp needs. Where the two routes use the same step count the ratio is that of their step costs; this is a property of the chosen operating points, not a dynamical equivalence of the two paths.


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[2] S. Bogner, H. Hergert, M. Hjorth-Jensen, R. LaRose, D. Lee, M. Patkowski, *Quantum state preparation with resolution refinement*, Phys. Lett. B **875**, 140363 (2026); arXiv:2511.14732.

[3] J. P. Sadia, M. Hjorth-Jensen, D. Lee, R. LaRose, *Digital quantum state preparation by nucleation*, arXiv:2609.37527 (2026).

TABLE S6. Convergence of the operating-point search: best CNOTs per accepted preparation found at the three search levels described in the text (L0: coarse grid of the first joint search; L1: fine $(T, \mu_{\mathrm{buf}})$ grid, geometric $n_s$ list; L2: L1 plus the integer refinement of $n_s$), the L2 operating point as $(T, n_s, M)$ or $(\mu_{\mathrm{buf}}, T, n_s, M)$, the infidelity reached there, the number of distinct controls within 10% of the L2 minimum, and the ratio refinement/ramp at L2. Filter settings and encoding as in Table V.

| route | target | L0 | L1 | L2 | L2 point | $1-\mathcal{F}$ | within 10% | ref./ramp |
|---|---|---|---|---|---|---|---|---|
| $k=4$, $N=4$, $g=1$ | | | | | | | | |
| HF ramp | $2\times10^{-3}$ | $9.47\times10^{3}$ | $6.51\times10^{3}$ | $6.51\times10^{3}$ | (4.25, 11, 0) | $1.7\times10^{-3}$ | 6 | |
| | $5\times10^{-4}$ | $1.89\times10^{4}$ | $1.07\times10^{4}$ | $9.47\times10^{3}$ | (4.5, 16, 0) | $5.0\times10^{-4}$ | 3 | |
| | $2\times10^{-4}$ | $2.55\times10^{4}$ | $1.24\times10^{4}$ | $1.24\times10^{4}$ | (4.75, 21, 0) | $1.8\times10^{-4}$ | 2 | |
| refinement $2\to4$ | $2\times10^{-3}$ | $9.92\times10^{3}$ | $6.82\times10^{3}$ | $6.20\times10^{3}$ | (2, 5, 10, 0) | $1.8\times10^{-3}$ | 3 | 0.95 |
| | $5\times10^{-4}$ | $1.36\times10^{4}$ | $8.06\times10^{3}$ | $8.06\times10^{3}$ | (2.5, 4.5, 13, 0) | $4.3\times10^{-4}$ | 2 | 0.85 |
| | $2\times10^{-4}$ | $1.98\times10^{4}$ | $1.12\times10^{4}$ | $1.12\times10^{4}$ | (2.5, 4.5, 18, 0) | $1.9\times10^{-4}$ | 1 | 0.90 |
| $k=8$, $N=8$, $g=1$ | | | | | | | | |
| HF ramp | $10^{-3}$ | $1.09\times10^{5}$ | $7.07\times10^{4}$ | $7.07\times10^{4}$ | (4.75, 26, 0) | $9.8\times10^{-4}$ | 4 | |
| | $3\times10^{-4}$ | $1.63\times10^{5}$ | $1.12\times10^{5}$ | $9.79\times10^{4}$ | (5.25, 36, 0) | $2.9\times10^{-4}$ | 3 | |
| | $10^{-4}$ | $2.18\times10^{5}$ | $1.52\times10^{5}$ | $1.36\times10^{5}$ | (5.25, 50, 0) | $9.5\times10^{-5}$ | 2 | |
| refinement $4\to8$ | $10^{-3}$ | $1.29\times10^{5}$ | $9.68\times10^{4}$ | $8.72\times10^{4}$ | (3, 5, 27, 0) | $8.9\times10^{-4}$ | 3 | 1.23 |
| | $3\times10^{-4}$ | $1.94\times10^{5}$ | $1.32\times10^{5}$ | $1.19\times10^{5}$ | (4, 5.5, 37, 0) | $2.8\times10^{-4}$ | 1 | 1.22 |
| | $10^{-4}$ | $2.58\times10^{5}$ | $2.10\times10^{5}$ | $1.94\times10^{5}$ | (4, 6.5, 60, 0) | $9.9\times10^{-5}$ | 3 | 1.42 |
| $k=8$, $N=8$, $g=2$ | | | | | | | | |
| HF ramp | $10^{-3}$ | $2.18\times10^{5}$ | $2.04\times10^{5}$ | $2.01\times10^{5}$ | (8, 74, 0) | $9.8\times10^{-4}$ | 2 | |
| | $3\times10^{-4}$ | $4.00\times10^{5}$ | $3.66\times10^{5}$ | $3.58\times10^{5}$ | (8, 98, 1) | $3.0\times10^{-4}$ | 4 | |
| | $10^{-4}$ | $5.83\times10^{5}$ | $5.49\times10^{5}$ | $5.29\times10^{5}$ | (8, 94, 3) | $9.9\times10^{-5}$ | 10 | |
| refinement $4\to8$ | $10^{-3}$ | $2.58\times10^{5}$ | $2.10\times10^{5}$ | $2.03\times10^{5}$ | (16, 7, 63, 0) | $9.8\times10^{-4}$ | 8 | 1.01 |
| | $3\times10^{-4}$ | $3.87\times10^{5}$ | $3.26\times10^{5}$ | $3.13\times10^{5}$ | (20, 7.5, 97, 0) | $2.9\times10^{-4}$ | 10 | 0.88 |
| | $10^{-4}$ | $5.70\times10^{5}$ | $4.69\times10^{5}$ | $4.55\times10^{5}$ | (24, 8.5, 141, 0) | $9.9\times10^{-5}$ | 13 | 0.86 |

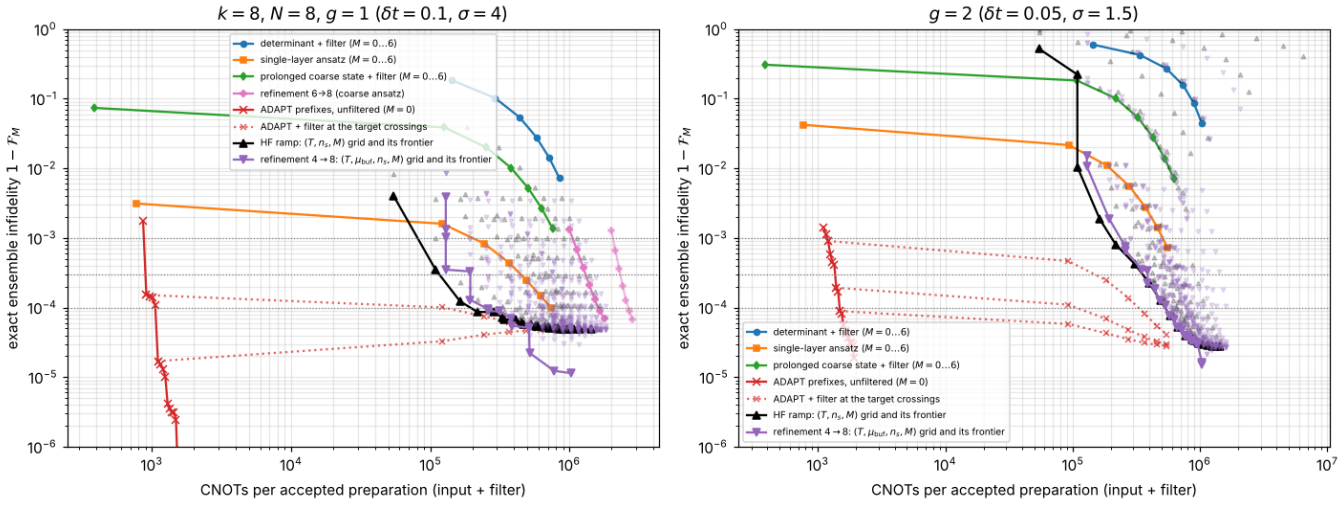



FIG. S9. The coarse (L0) control grid of the first joint search at $k=8$, $N=8$, with the exact averaged channel of the digital filter: ensemble infidelity of the accepted runs against CNOTs per accepted preparation, Eq. (12), at $g=1$ (left; $\delta t=0.1$, $\sigma=4$) and $g=2$ (right; $\delta t=0.05$, $\sigma=1.5$). Lines with markers: the fixed inputs of Table VI as $M$ grows from 0 to 6, the ADAPT prefixes unfiltered (crosses) and the three prefixes at the target crossings through the filter (dotted); faint symbols: every point of the $(T, n_s, M)$ grid of the ramp and of the $(\mu_{\mathrm{buf}}, T, n_s, M)$ grid of the $4\to8$ refinement, with their lower envelopes drawn (the L0 minima of Table S6). Dotted lines mark the targets $10^{-3}$, $3\times10^{-4}$ and $10^{-4}$.